

7:42

Tarea

$$\lim_{x \rightarrow 0} \frac{\sqrt{4+x} - 2}{\sqrt{9+x} - 3}$$

$$\lim_{x \rightarrow 2} \frac{x^4 - 8x^2 + 16}{x - 2}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{4+x} - 2}{\sqrt{9+x} - 3}$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{4+x} - 2) \cdot (\sqrt{4+x} + 2)}{(\sqrt{9+x} - 3) \cdot (\sqrt{9+x} + 3)}$$

$$\lim_{x \rightarrow 0} \frac{x + x - 4}{(\sqrt{9+x} - 3) \cdot (\sqrt{9+x} + 3)}$$

$$\lim_{x \rightarrow 0} \frac{x}{(\sqrt{9+x} - 3)(\sqrt{9+x} + 3)} \cdot \frac{(\sqrt{9+x} + 3)}{(\sqrt{9+x} + 3)}$$

$$\lim_{x \rightarrow 0} \frac{x(\sqrt{9+x} + 3)}{(\sqrt{9+x} + 2)(x + x - 4)} = \lim_{x \rightarrow 0} \frac{x(\sqrt{9+x} + 3)}{(\sqrt{9+x} + 2) \cdot x}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{9+x} + 3}{\sqrt{9+x} + 2} = \frac{\sqrt{9+0} + 3}{\sqrt{9+0} + 2} = \frac{3+3}{2+2} = \frac{6}{4} = \frac{3}{2}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{4+x} - 2}{\sqrt{9+x} - 3} = \frac{3}{2}$$